Week 10: 2nd – 6th 2026

Subject / الموضوع: Science **Unit Title / عنوان الوحدة: Forces, Motion and Energy** **Grade / الصف: 8**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Session 1 الحصة الأولى **Position and Motion** الموقع والحركة	**DOK 3 – Analyze:** I can analyze data and distance–time graphs to describe an object's position and motion from a reference point, and calculate and compare speeds.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.1 Activate prior knowledge using organizers, links, and pre-teaching. 5.3 Build fluency with graduated scaffolds and models.	Kindly check Toddle for the lesson PPT and practice sheet.
Session 2 الحصة الثانية **Force and Acceleration / Force Pairs** القوة والتسارع / أزواج القوى	**DOK 3 – Explain with evidence:** I can use evidence from an investigation to explain how net force and mass affect an object's acceleration, and identify the force pairs acting in real situations.	3.3 Guide processing with explicit steps, chunking, and interactive models. 8.3 Foster collaboration and community with structured group roles and supports. 4.1 Provide alternatives for physical action and methods of response.	Kindly check Toddle for the lesson PPT and practice sheet.
Session 3 الحصة الثالثة **Gravitational Force** قوة الجاذبية	**DOK 3 – Analyze:** I can analyze how mass and distance affect the strength of the gravitational force between objects, and use evidence to explain the difference between mass and weight.	7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Kindly check Toddle for the lesson PPT and practice sheet.
Session 4 الحصة الرابعة **Kinetic and Potential Energy** الطاقة الحركية وطاقة الوضع	**DOK 3 – Analyze data:** I can analyze data to explain how kinetic energy depends on mass and speed and how potential energy depends on height, and describe how energy changes from one form to the other.	7.1 Increase engagement by offering choices and autonomy. 5.1 Provide multiple communication media (text, speech, visuals, models). 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	Kindly check Toddle for the lesson PPT and practice sheet.

Homework **الواجبات**			
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**			

Week 10 الأسبوع 10

Technology Integration دمج التكنولوجيا	Videos and online simulations of motion, forces and energy, and Toddle for practice sheets and feedback.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics (rearranging formulas, graphs and units) and Physical and Health Education (forces and energy in sport).
External Assessment Integration دمج التقييمات الخارجية	Aligned to NGSS MS-PS2-1, MS-PS2-2, MS-PS2-4, MS-PS3-1 and MS-PS3-2, and MAP Growth Science (Physical Sciences).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking (analysing data and graphs), collaboration (group investigations) and communication (explaining answers with scientific vocabulary and units).
Standards المعايير	• **MS-PS2-1:** Apply Newton's third law to design a solution to a problem involving the motion of two colliding objects. • **MS-PS2-2:** Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object. • **MS-PS2-4:** Construct and present arguments using evidence to support the claim that gravitational interactions are attractive and depend on the masses of interacting objects. • **MS-PS3-1:** Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass of an object and to the speed of an object. • **MS-PS3-2:** Develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts of potential energy are stored in the system.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Inquiry:** How do forces change the motion of objects, and how is energy transferred when objects move? **TOK:** If motion always depends on a reference point, can any measurement of motion be completely objective?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Session 1 – Position and Motion** • **MUST (DOK 1):** I can define position, reference point, distance, displacement and speed. • **SHOULD (DOK 2):** I can calculate speed using speed = distance ÷ time with the correct units, and describe motion shown on a distance–time graph. • **COULD (DOK 3):** I can compare the motion of two objects on the same distance–time graph and explain where and why their speeds change. **Session 2 – Force and Acceleration / Force Pairs** • **MUST (DOK 1):** I can define force, net force, balanced and unbalanced forces, and acceleration, and identify an action–reaction force pair. • **SHOULD (DOK 2):** I can use F = m × a to calculate force, mass or acceleration, and explain Newton's third law with an example.

	• **COULD (DOK 3):** I can predict how changing the force or the mass changes the acceleration, and justify why a force pair does not cancel out. **Session 3 – Gravitational Force** • **MUST (DOK 1):** I can define gravitational force, mass and weight, and state that mass is measured in kg and weight in N. • **SHOULD (DOK 2):** I can calculate weight using W = m × g and explain how mass and distance change the gravitational force. • **COULD (DOK 3):** I can compare an object's weight on Earth, the Moon and Mars, and justify the differences using data. **Session 4 – Kinetic and Potential Energy** • **MUST (DOK 1):** I can define kinetic energy and gravitational potential energy and identify examples of each. • **SHOULD (DOK 2):** I can calculate KE = ½ × m × v^2 and GPE = m × g × h, and explain the energy changes on a roller coaster. • **COULD (DOK 3):** I can use data to justify why speed has a greater effect on kinetic energy than mass, and analyze the energy changes in a real system.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء Global Understanding التفاهم العالمي Arabic Language اللغة العربية
	المجالArea: Pride in UAE achievements in science, transport and space exploration.
	التطبيقImplementation: • Session 1: students calculate and compare the speed of the Dubai Metro between stations using journey data. • Session 3: students compare weight on Earth, the Moon and Mars, linked to the Emirates Mars Mission (Hope Probe) and the Rashid lunar rover. • Session 4: students analyze the energy changes on Formula Rossa at Ferrari World Abu Dhabi. • Arabic key terms: السرعة (speed), القوة (force), التسارع (acceleration), الجاذبية (gravity), الطاقة الحركية (kinetic energy), طاقة الوضع (potential energy).